

Software Engineering

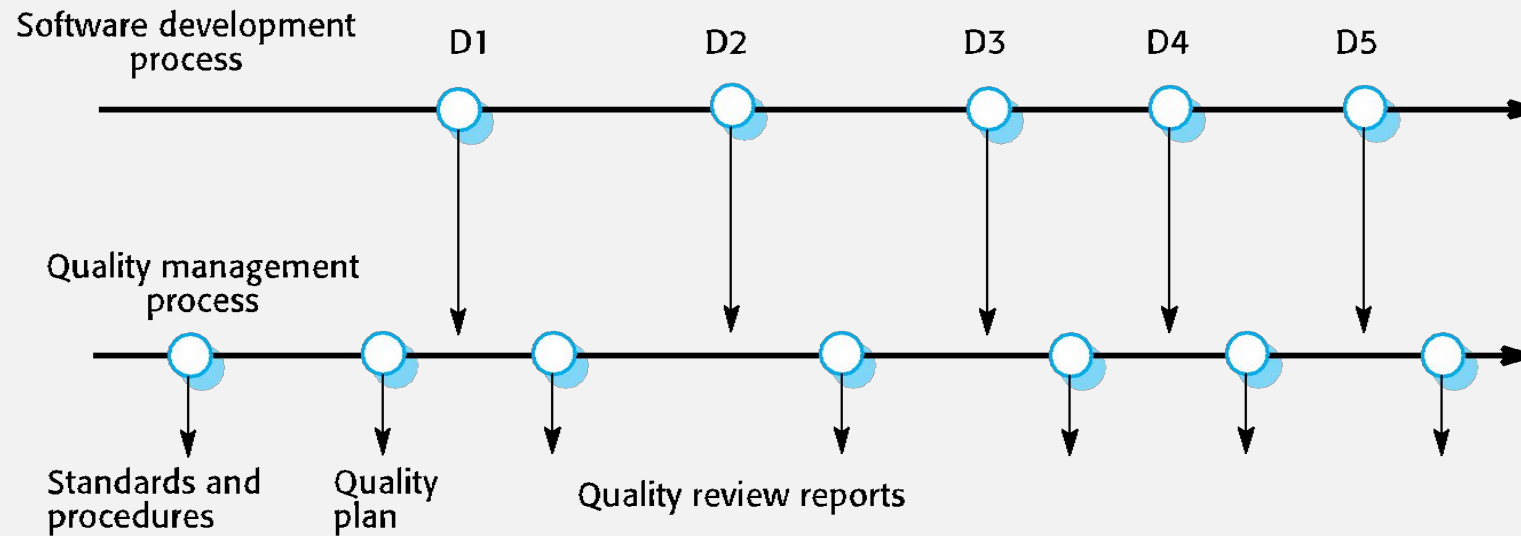
Quality management

- Software quality
- Software standards
- Reviews and inspections
- Quality management and agile development
- Software measurement

Quality management activities

- Quality management provides an independent check on the software development process.
- The quality management process checks the project deliverables to ensure that they are consistent with organizational standards and goals
- The quality team should be independent from the development team so that they can take an objective view of the software. This allows them to report on software quality without being influenced by software development issues.

Quality management and software development



Software quality attributes

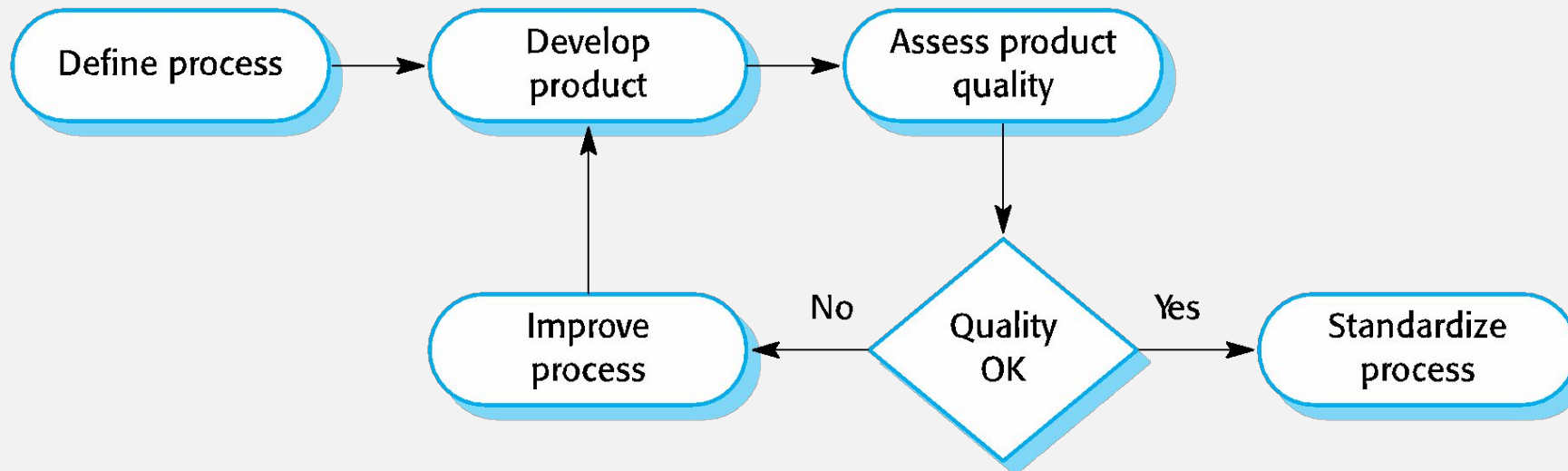
Safety	Understandability	Portability
Security	Testability	Usability
Reliability	Adaptability	Reusability
Resilience	Modularity	Efficiency
Robustness	Complexity	Learnability

The subjective quality of a software system is largely based on its non-functional characteristics.

Quality conflicts

- It is not possible for any system to be optimized for all of these attributes – for example, improving robustness may lead to loss of performance.
- The quality plan should therefore define the most important quality attributes for the software that is being developed.
- The plan should also include a definition of the quality assessment process, an agreed way of assessing whether some quality, such as maintainability or robustness, is present in the product.

Process-based quality



Importance of standards

- Encapsulation of best practice- avoids repetition of past mistakes.
- They are a framework for defining what quality means in a particular setting i.e. that organization's view of quality.
- They provide continuity - new staff can understand the organisation by understanding the standards that are used.

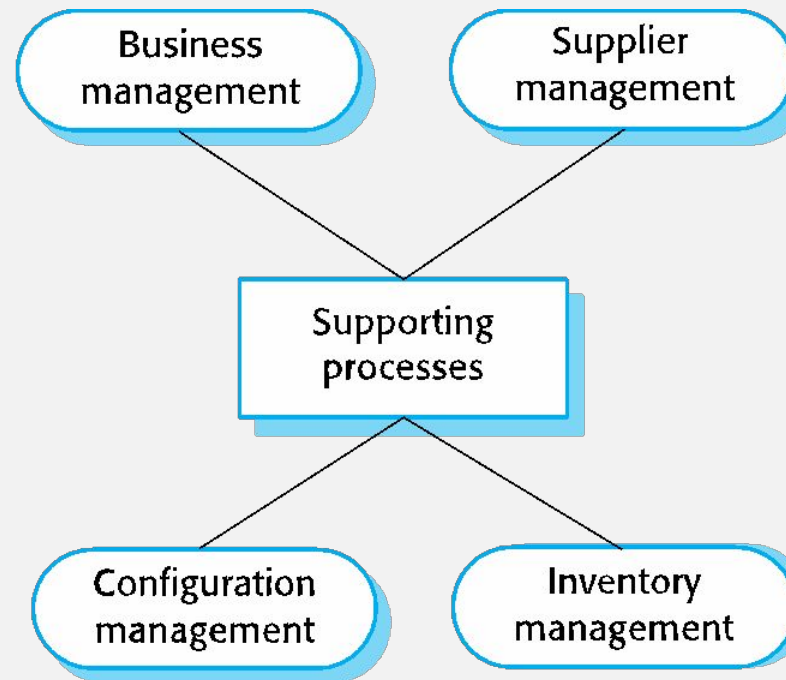
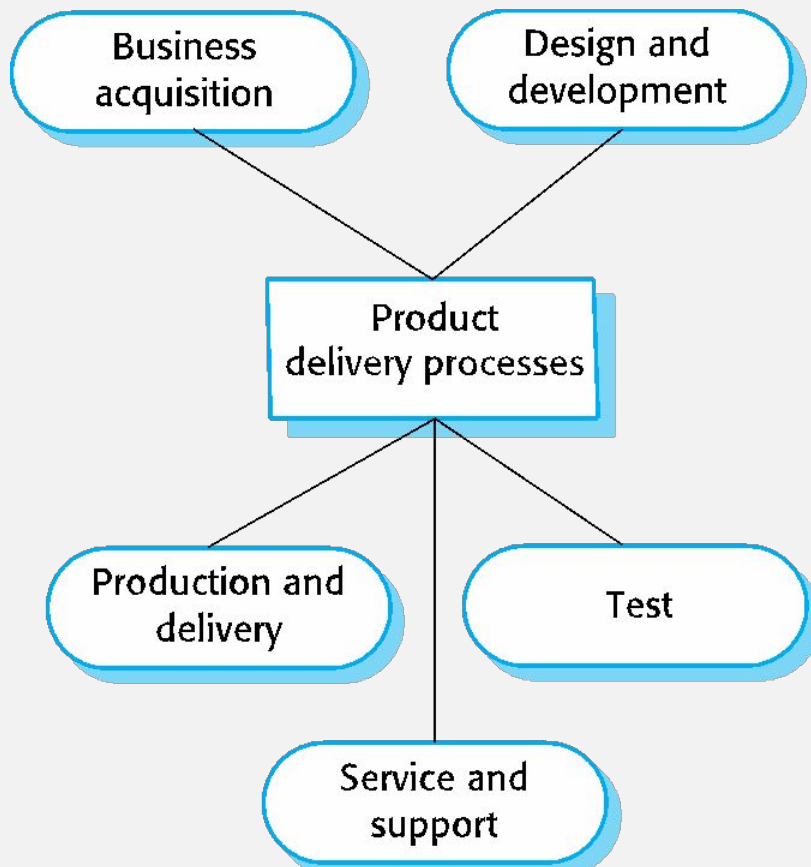
Software standards

Standards define the required attributes of a product or process. They play an important role in quality management. Standards may be international, national, organizational or project standards.

Product standards	Process standards
Design review form	Design review conduct
Requirements document structure	Submission of new code for system building
Method header format	Version release process
Java programming style	Project plan approval process
Project plan format	Change control process
Change request form	Test recording process

- *Product standards*
 - Apply to the software product being developed. They include document standards, such as the structure of requirements documents, documentation standards, such as a standard comment header for an object class definition, and coding standards, which define how a programming language should be used.
- *Process standards*
 - These define the processes that should be followed during software development. Process standards may include definitions of specification, design and validation processes, process support tools and a description of the documents that should be written during these processes.

ISO 9001 core processes

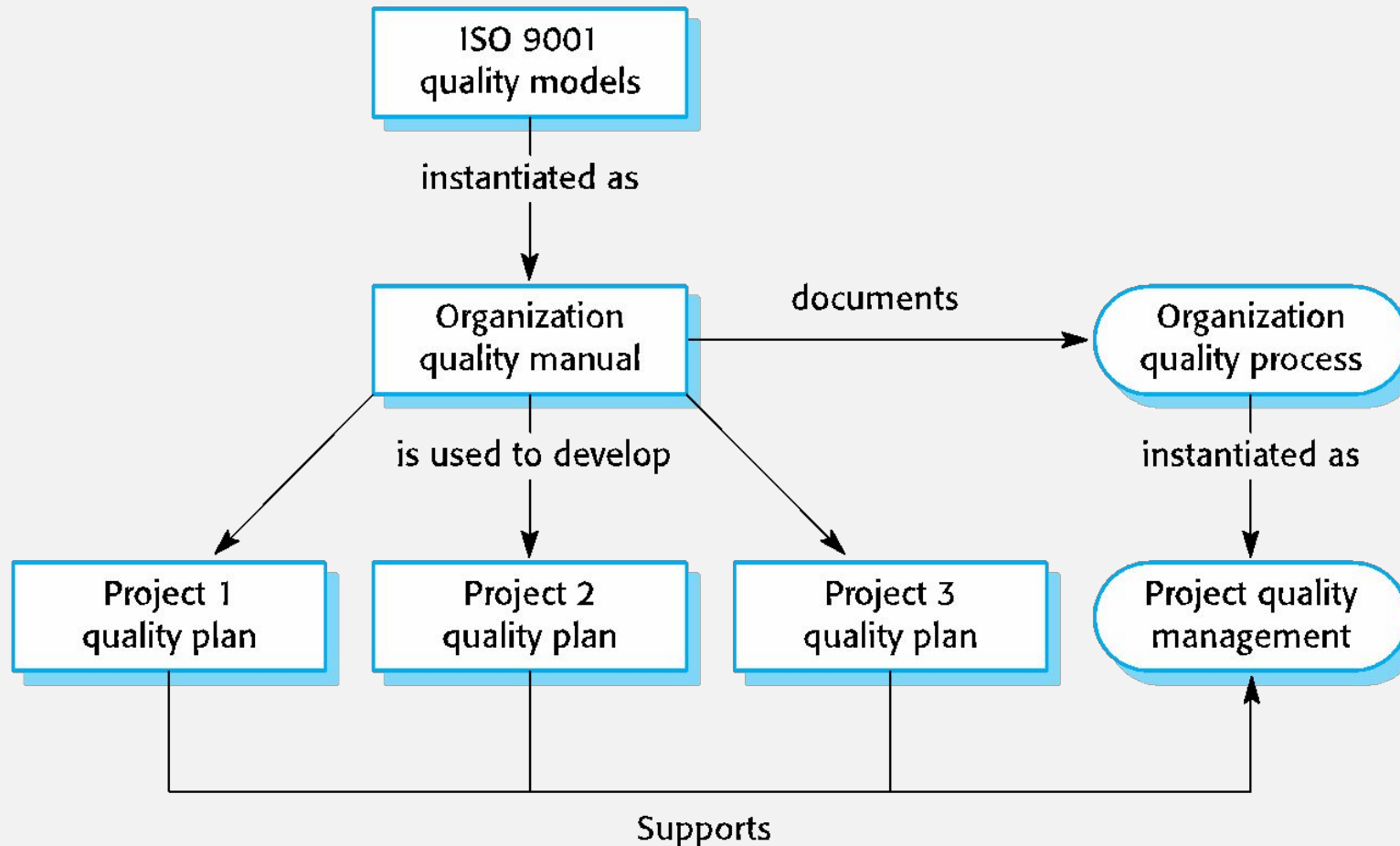


An international set of standards that can be used as a basis for developing quality management systems.

ISO 9001, the most general of these standards, applies to organizations that design, develop and maintain products, including software.

The ISO 9001 standard is a framework for developing software standards.

ISO 9001 and quality management



ISO 9001 certification

- Quality standards and procedures should be documented in an organisational quality manual.
- An external body may certify that an organisation's quality manual conforms to ISO 9000 standards.
- Some customers require suppliers to be ISO 9000 certified although the need for flexibility here is increasingly recognised.



Estimation for Software Projects

Lines of Code and Functional Points

Software Project Estimating

- Estimation of the size of software is an essential part of Software Project Management. It helps the project manager to further predict the effort and time which will be needed to build the project.
- Various measures are used in project size estimation. Some of these are:
 - LOC
 - Functional Point

Software Project Estimating

- LOC and FP data are used in two ways during software project estimation:
 - (1) as estimation variables to “size” each element of the software
 - (2) as baseline metrics collected from past projects and used in conjunction with estimation variables to develop cost and effort projections.

LOC and FP estimation are distinct estimation techniques. Yet both have a number of characteristics in common.

Baseline productivity metrics (e.g., LOC/pm or FP/pm)

LOC- with historical data

A three-point or expected value can then be computed. The *expected value* for the estimation variable (size) S can be computed as a weighted average of the optimistic (s_{opt}), most likely (s_m), and pessimistic (s_{pess}) estimates. For example,

$$S = \frac{s_{\text{opt}} + 4s_m + s_{\text{pess}}}{6} \quad (26.1)$$

- Once the expected value for the estimation variable has been determined, historical LOC or FP productivity data are applied.

Example of LOC

- The mechanical CAD software will accept two- and three-dimensional geometric data from an engineer. The engineer will interact and control the CAD system through a user interface that will exhibit characteristics of good human/machine interface design. All geometric data and other supporting information will be maintained in a CAD database.
- Design analysis modules will be developed to produce the required output, which will be displayed on a variety of graphics devices. The software will be designed to control and interact with peripheral devices that include a mouse, digitizer, laser printer, and plotter.

Example Continue

- A range of LOC estimates is developed for each function.
- For example, the range of LOC estimates for the 3D geometric analysis function is
 1. optimistic, 4600 LOC;
 2. most likely, 6900 LOC;
 3. pessimistic, 8600 LOC.

Applying Equation 26.1, the expected value for the 3D geometric analysis function is 6800 LOC.

$$S = \frac{S_{\text{opt}} + 4S_m + S_{\text{pess}}}{6}$$

Function	Estimated LOC
User interface and control facilities (UICF)	2,300
Two-dimensional geometric analysis (2DGA)	5,300
Three-dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (CGDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	33,200

A review of historical data indicates that the organizational average productivity for systems of this type is 620 LOC/pm. And Based on a burdened labor rate of \$8000 per).

1. No of pm = $33200/620 = 54$ person month.
2. No of LOC/pm = $33200/54 = 626$ loc/pm.
3. Cost per line = $\$8000/626 = \13
4. Total Cost Project = $8000*54 = \$ 431,000$

Function Point

- In this method, the number and type of functions supported by the software are utilized to find FPC(function point count). The steps in function point analysis are:
 1. Count the number of functions of each proposed type. **count**
 2. Compute the Unadjusted Function Points(UFP).
 3. Find Total Degree of Influence(TDI).
 4. Compute Value Adjustment Factor(VAF).
 5. Find the Function Point Count(FPC).

Count the number of functions of each proposed type

- Find the number of functions belonging to the following types:
 1. External Inputs: Functions related to data entering the system.
 2. External outputs: Functions related to data exiting the system.
 3. External Inquiries: They leads to data retrieval from system but don't change the system.
 4. Internal Files: Logical files maintained within the system. Log files are not included here.
 5. External interface Files: These are logical files for other applications which are used by our system.

Information Domain Value	Count		Weighting factor				
			Simple	Average	Complex		
External Inputs (EIs)		×	3	4	6	=	
External Outputs (EOs)		×	4	5	7	=	
External Inquiries (EQs)		×	3	4	6	=	
Internal Logical Files (ILFs)		×	7	10	15	=	
External Interface Files (EIFs)		×	5	7	10	=	
Count total	→						

To compute function points (FP), the following relationship is used:

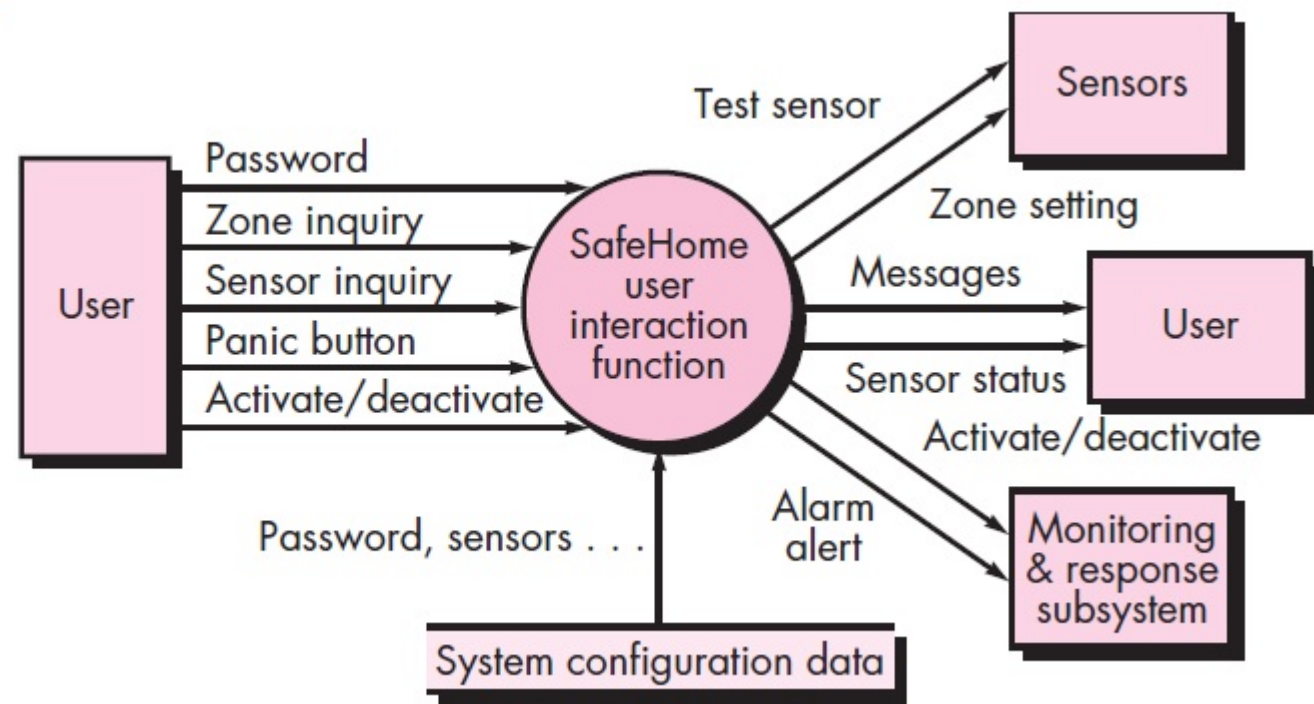
$$FP = \text{count total} \times [0.65] + 0.01 \times \sum (F_i)$$

Value adjustment factors are used to provide an indication of problem complexity.

1. Does the system require reliable backup and recovery?
2. Are specialized data communications required to transfer information to or from the application?
3. Are there distributed processing functions?
4. Is performance critical?
5. Will the system run in an existing, heavily utilized operational environment?
6. Does the system require online data entry?
7. Does the online data entry require the input transaction to be built over multiple screens or operations?
8. Are the ILFs updated online?
9. Are the inputs, outputs, files, or inquiries complex?
10. Is the internal processing complex?
11. Is the code designed to be reusable?
12. Are conversion and installation included in the design?
13. Is the system designed for multiple installations in different organizations?
14. Is the application designed to facilitate change and ease of use by the user?

0 (not important or applicable) to 5 (absolutely essential)

A data flow model for *SafeHome* software



The data flow diagram is evaluated to determine a set of key information domain measures required for computation of the function point metric.

- Three external inputs—password, panic button, and activate/deactivate.**
- Two external inquiries—zone inquiry and sensor inquiry.**
- One ILF (system configuration file)**
- Two external outputs (messages and sensor status)**
- Four EIFs (test sensor, zone setting, activate/deactivate, and alarm alert)**

The count total shown in Figure 23.3 must be adjusted using Equation (23.1). For the purposes of this example, we assume that $\Sigma(F_i)$ is 46 (a moderately complex product). Therefore,

$$FP = 50 \times [0.65 + (0.01 \times 46)] = 56$$

Based on the projected FP value derived from the requirements model, the project team can estimate the overall implemented size of the *SafeHome* user interaction

Information Domain Value	Count		Weighting factor				
			Simple	Average	Complex		
External Inputs (EIs)	3	×	3	4	6	=	9
External Outputs (EOs)	2	×	4	5	7	=	8
External Inquiries (EQs)	2	×	3	4	6	=	6
Internal Logical Files (ILFs)	1	×	7	10	15	=	7
External Interface Files (EIFs)	4	×	5	7	10	=	20
Count total							50

- Assume that past data indicates that one FP translates into 60 lines of code (an object-oriented language is to be used) and that 12 FPs are produced for each person-month of effort. These historical data provide the project manager with important planning information that is based on the requirements model rather than preliminary estimates. Assume further that past projects have found an average of three errors per function point during requirements and design reviews and four errors per function point during unit and integration testing. These data can ultimately help you assess the completeness of your review and testing activities.

Information domain value	Opt.	Likely	Pess.	Est. count	Weight	FP count
Number of external inputs	20	24	30	24	4	97
Number of external outputs	12	15	22	16	5	78
Number of external inquiries	16	22	28	22	5	88
Number of internal logical files	4	4	5	4	10	42
Number of external interface files	2	2	3	2	7	15
<i>Count total</i>						320

Factor	Value
Backup and recovery	4
Data communications	2
Distributed processing	0
Performance critical	4
Existing operating environment	3
Online data entry	4
Input transaction over multiple screens	5
Master files updated online	3
Information domain values complex	5
Internal processing complex	5
Code designed for reuse	4
Conversion/installation in design	3
Multiple installations	5
Application designed for change	5
Value adjustment factor	1.17

$$FP = \text{Count-total} * [0.65 + 0.01 * \sum(f_i)]$$

$$320 * [0.65 + 0.01 * 52]$$

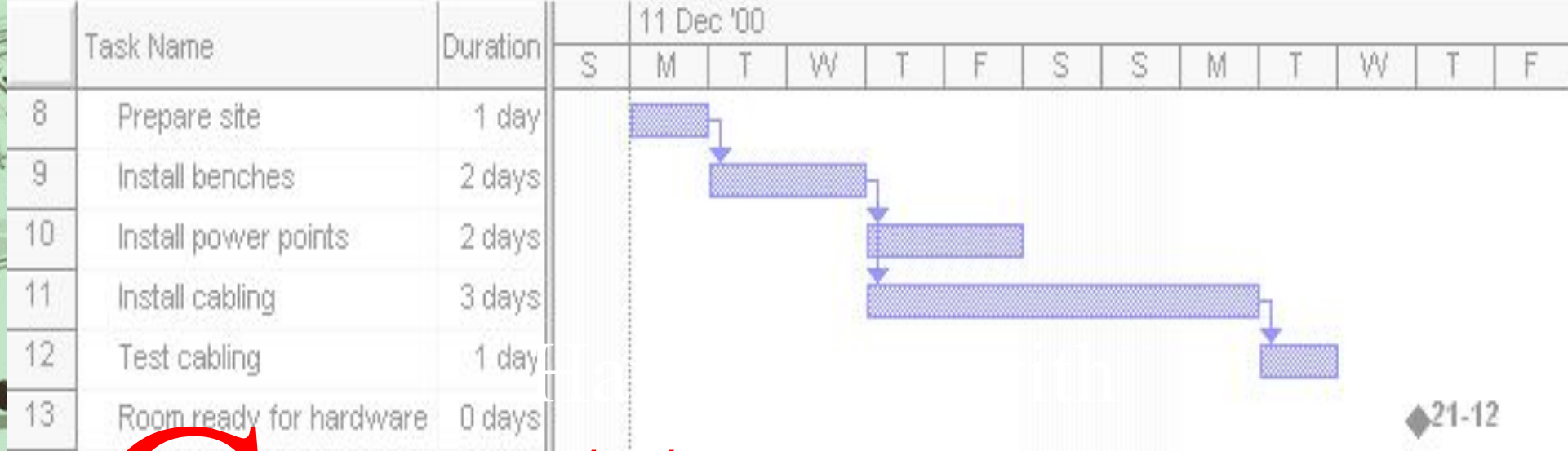
$$FP = 320 * [1.17]$$

**Given*

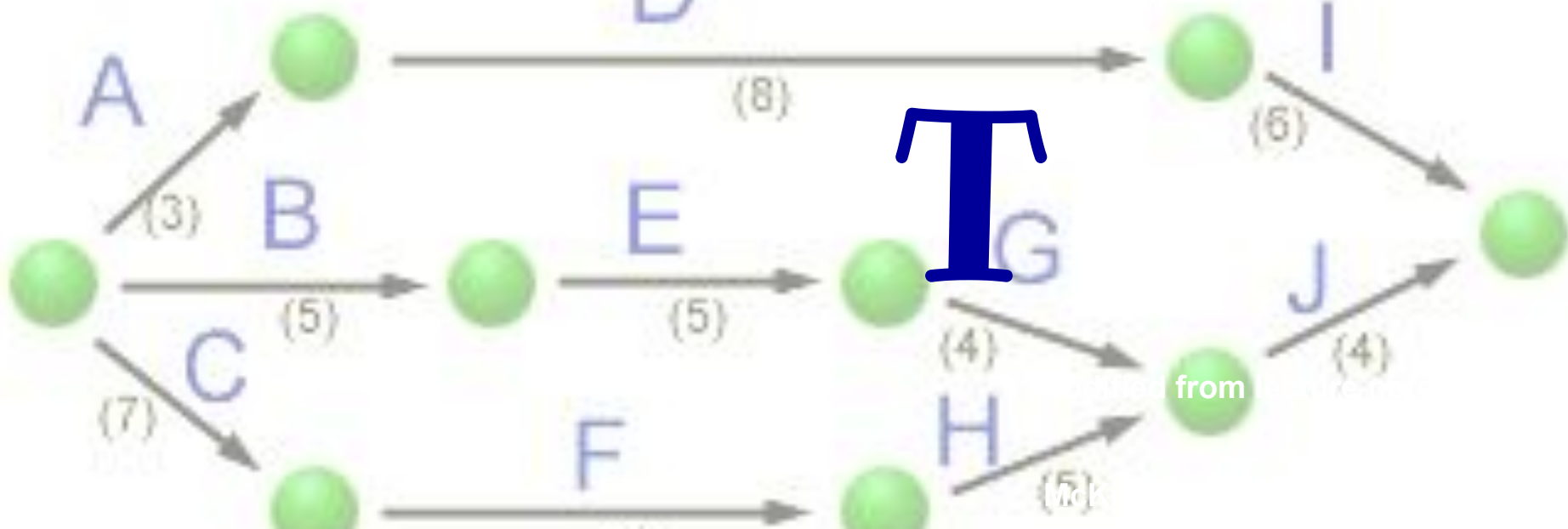
$$FP = 375$$

The organizational average productivity for systems of this type is 6.5 FP/pm.

- **Cost Per FP = 8000/6.5**
Based on a burdened labor rate of \$8000 per month, the cost per FP is approximately \$1230.
- **Project Cost = 1230*375 = 461000\$**
Based on the FP estimate and the historical productivity data, the total estimated project cost is \$461,000
- **Effort= FP/ 6.5 FP/pm= 375/6.5=58**
Estimated effort is 58 person-months.



Gantt & PERT

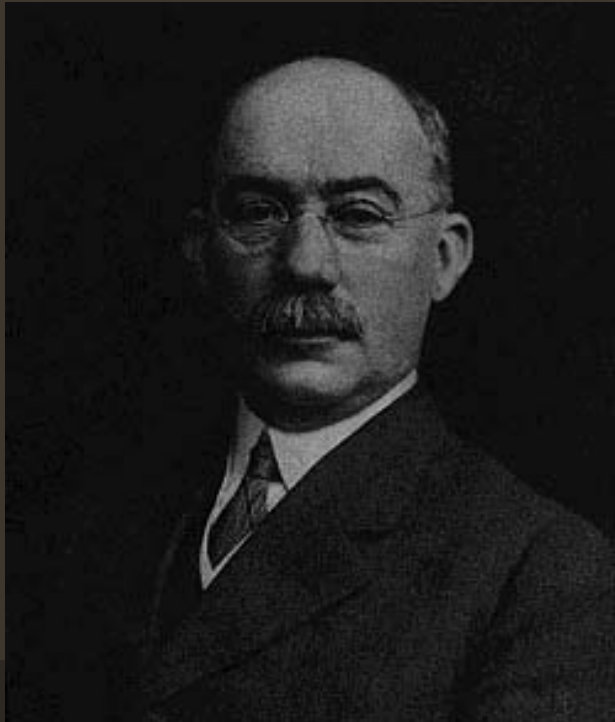


What are they?

Gantt and PERT charts are both “CPM” (Critical Path Method) tools to:

- manage the tasks involved in big and complex projects
- let project managers organise time, people, equipment and money
- ensure the right people and equipment are in the right place and the right time
- allow managers to monitor the progress of a project

Gantt Charts

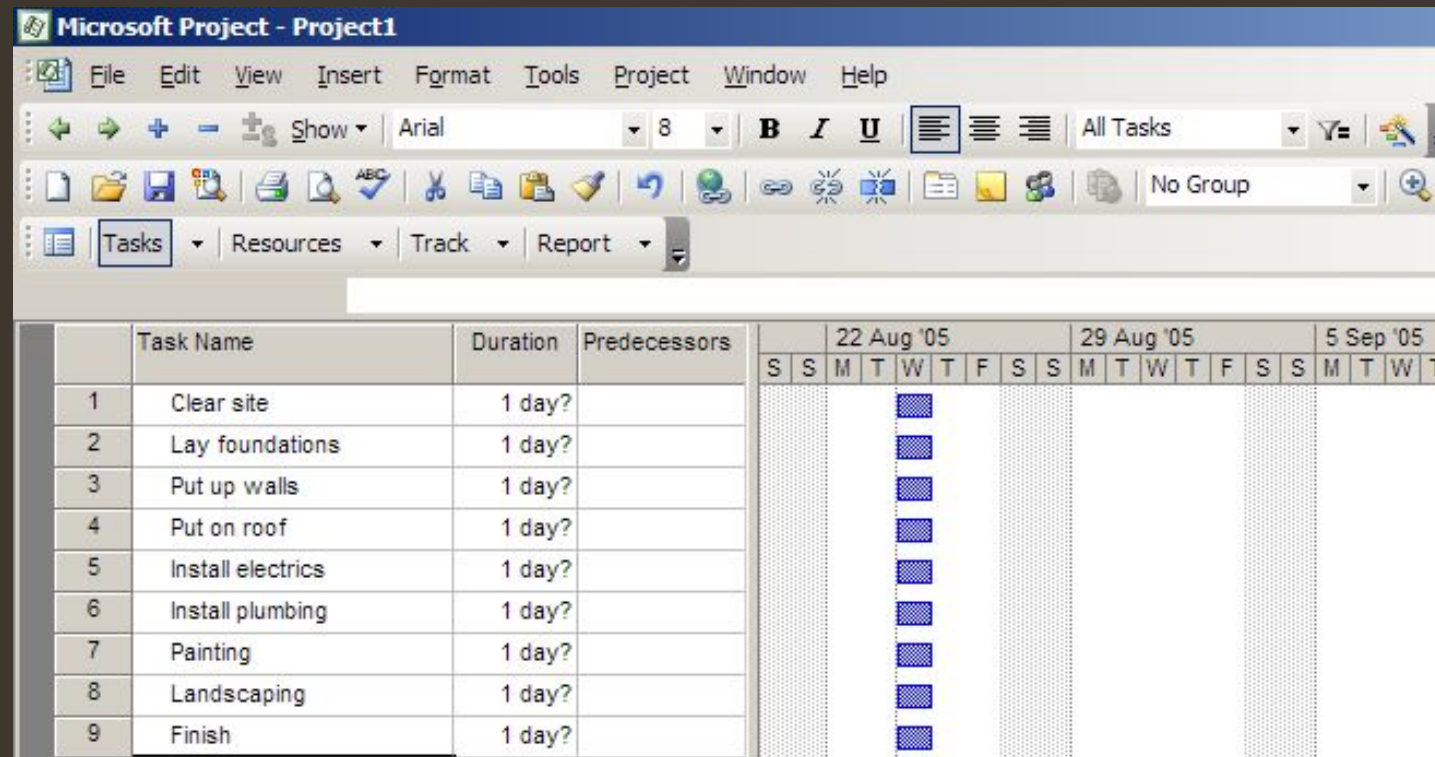


Gantt Basics

- Basically, a timeline with tasks that can be connected to each other
- Note the spelling!
- It is not all-capitals!
- Can be created with simple tools like Excel, but specialised tools like Microsoft Project make life easier

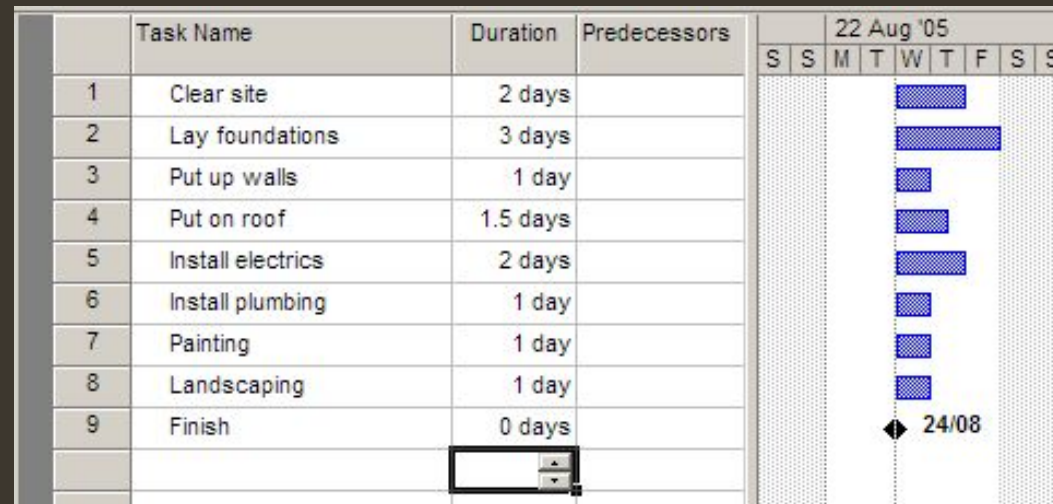
Making a Gantt chart

- Step 1 – list the tasks in the project



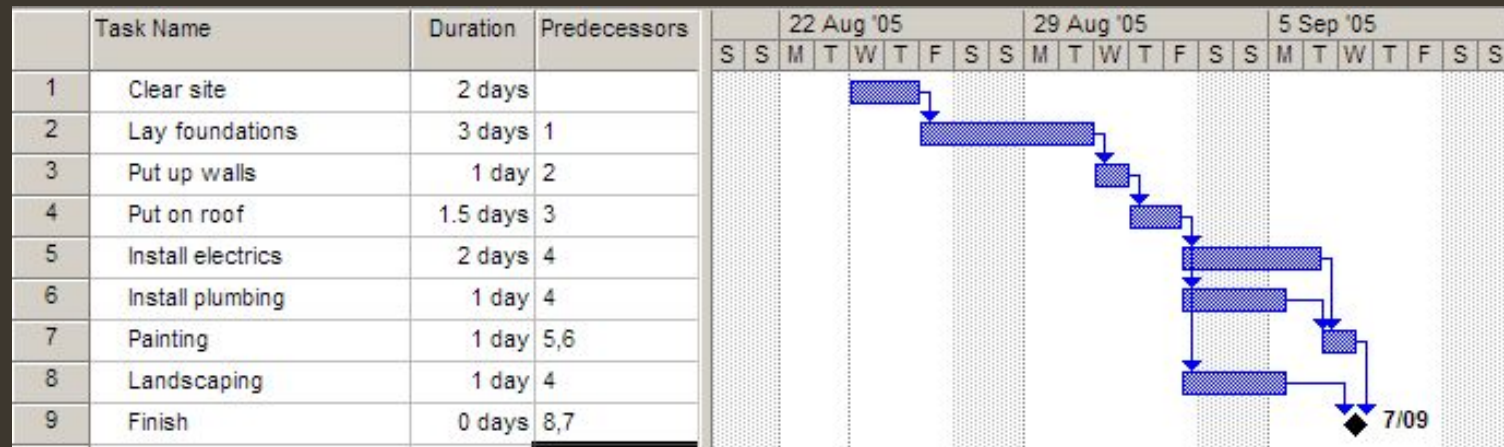
Making a Gantt chart

- Step 2 – add task durations

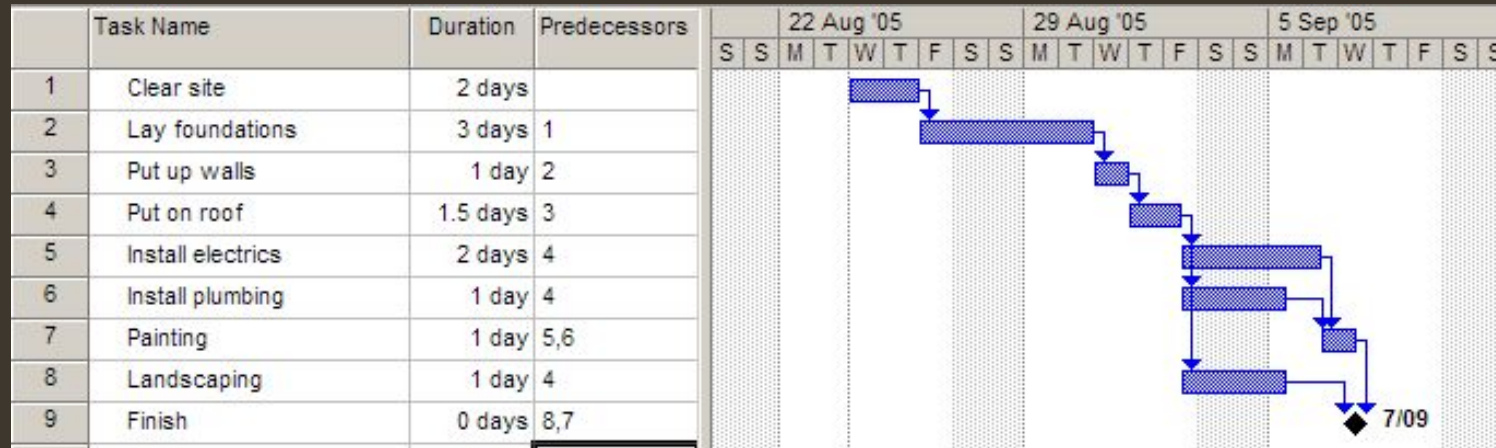


Making a Gantt chart

Step 3 – add *dependencies* (which tasks cannot start before another task finishes)



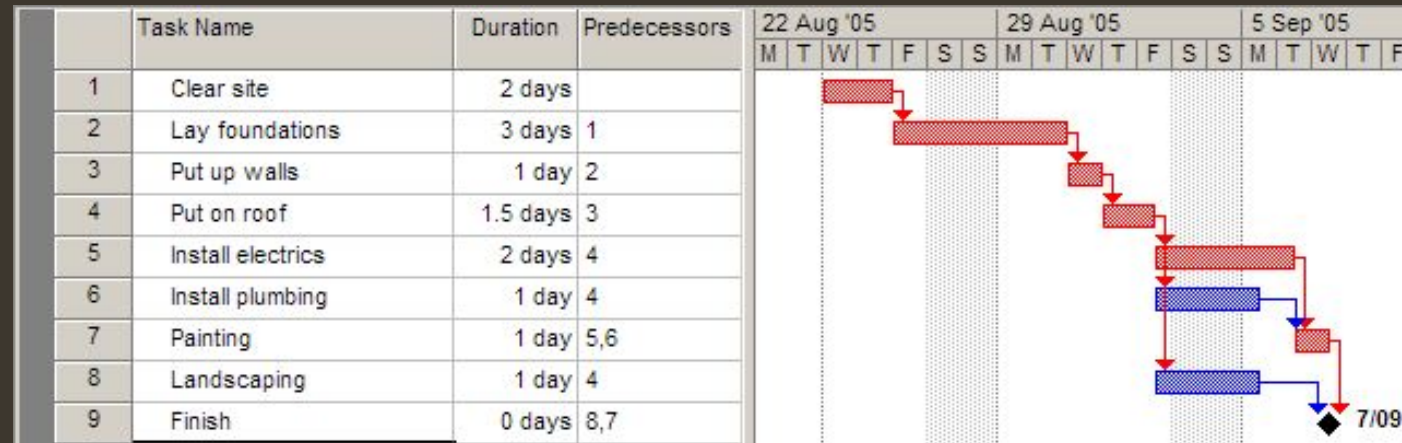
Notes



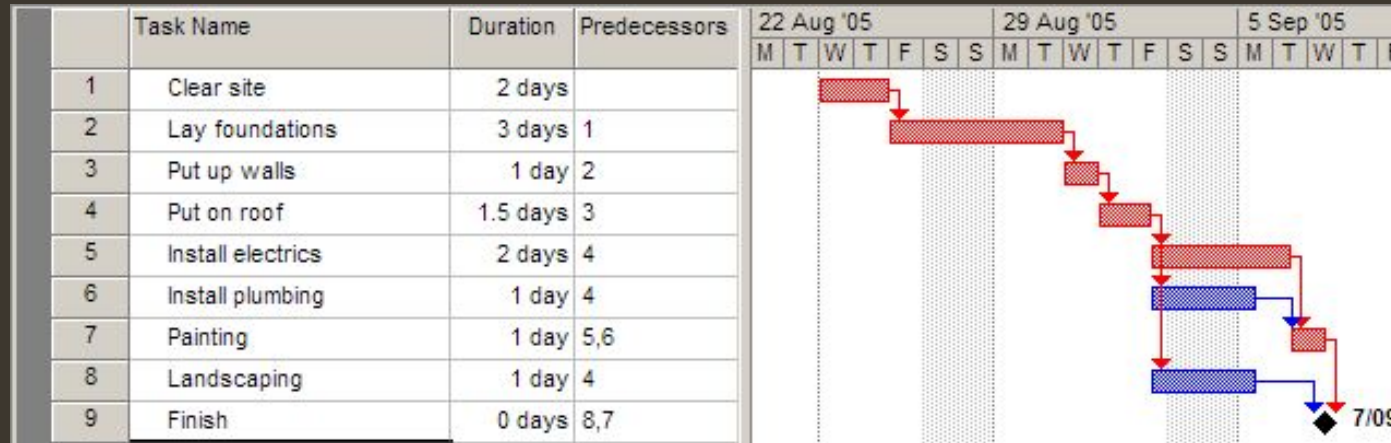
- The arrows indicate **dependencies**.
- Task 1 is a **predecessor** of task 2 – i.e. task 2 cannot start before task 1 ends.
- Task 3 is **dependent** on task 2. Task 7 is dependent on two other tasks
- Electrics, plumbing and landscaping are **concurrent** tasks and can happen at the same time, so they overlap on the chart. All 3 can start after task 4 ends.
- Painting must wait for both electrics and plumbing to be finished.
- Task 9 has zero duration, and is a **milestone**

Making a Gantt chart

- Step 4 – find the **critical path**

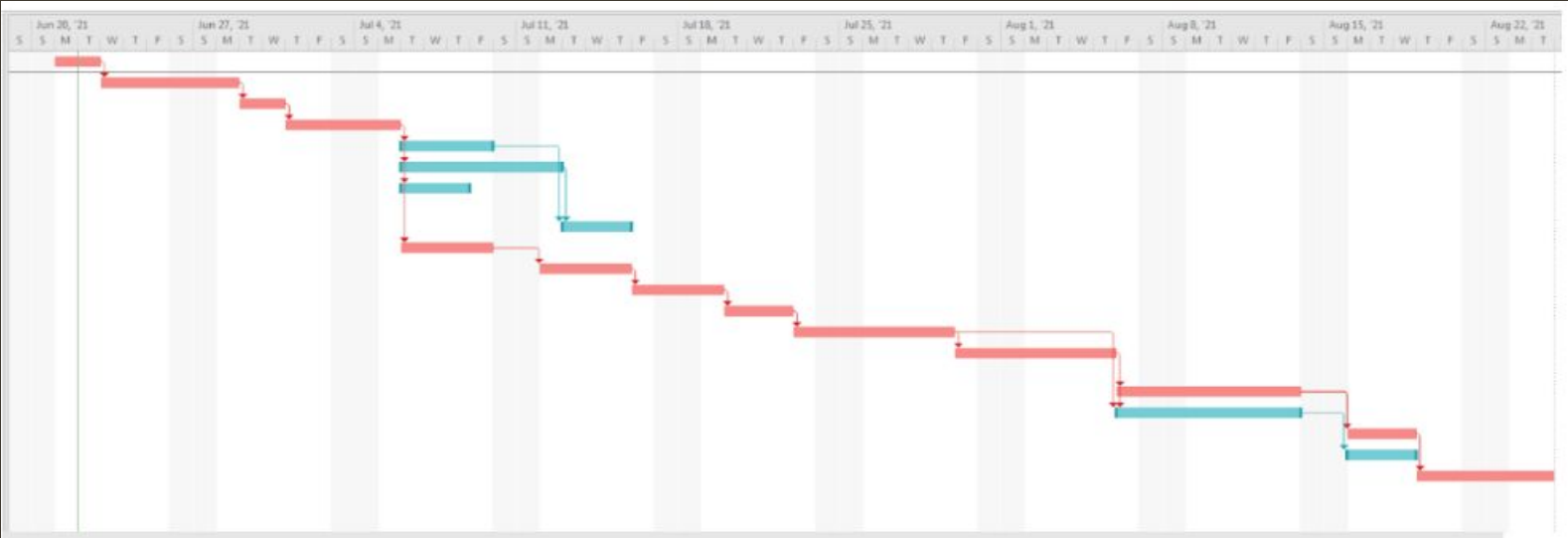
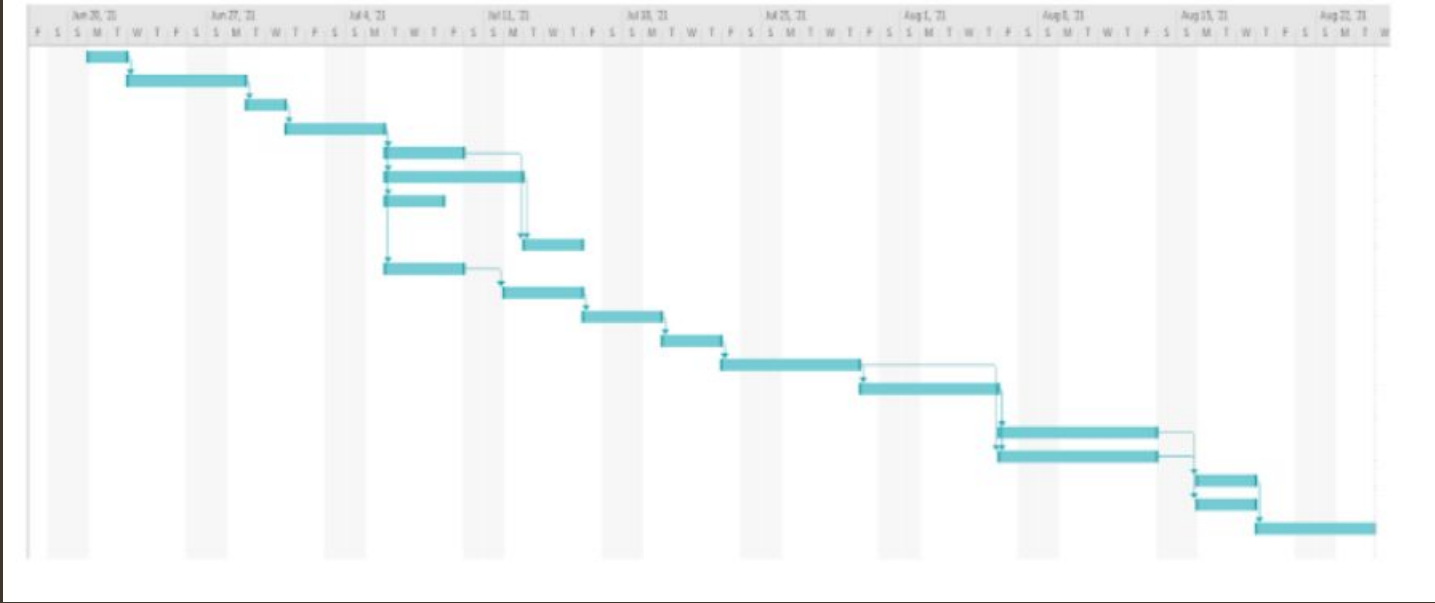


The **critical path** contains a list of linked tasks that directly affects the **project** finish date. If any task on the **critical path** is late, the whole **project** is late



- MS Project can work out the critical path for you!
- The length of the critical path is the sum of the lengths of all critical tasks (the red tasks 1,2,3,4,5,7) which is $2+3+1+1.5+2+1 = 10.5$ days.
- In other words, the minimum amount of time required to get all tasks completed is 10.5 days
- The other tasks (6,8) can each run over-time before affecting the end date of the project

example

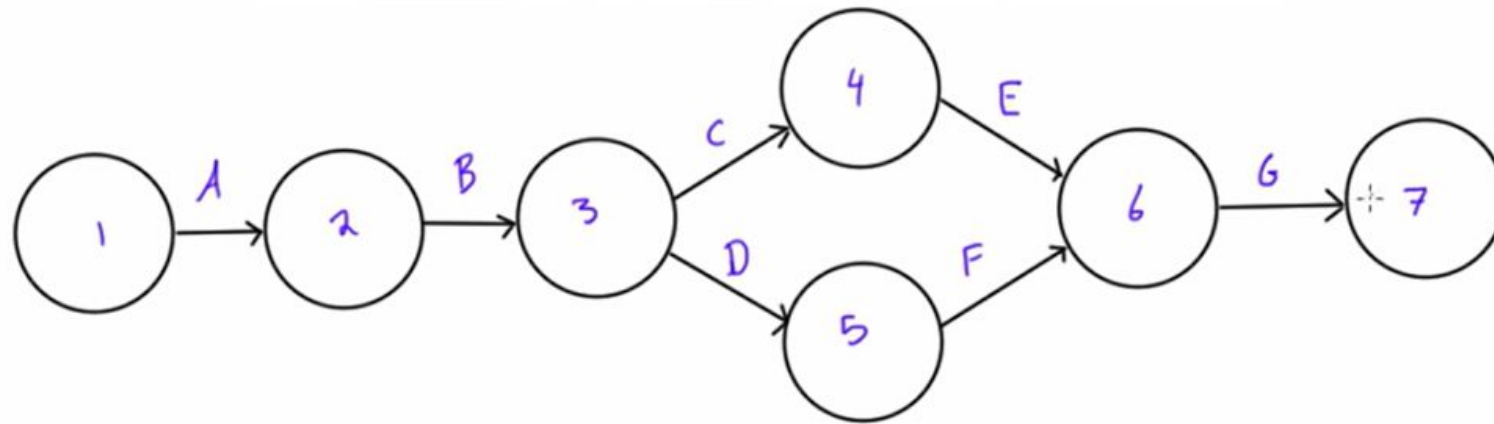


PERT basics

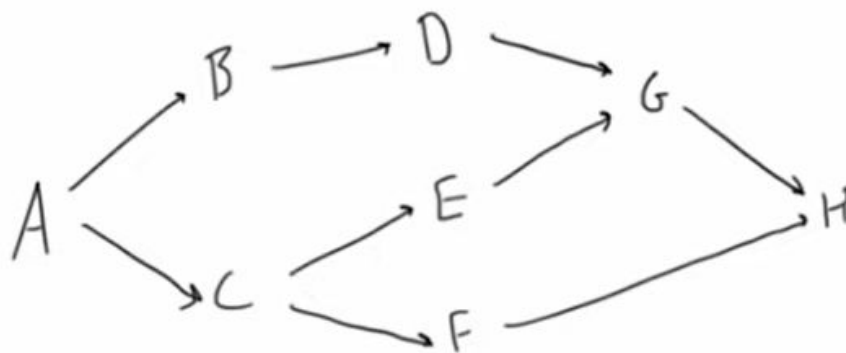
- PERT is an acronym so it's in capital letters
- Gantt is a name, so only has an initial capital
- In Gantt chart, the length of a task's bar is proportional to the length of the task. This rarely applies to PERT charts.
- There are a few different “flavours” of PERT and Gantt charts...

Pert Diagram

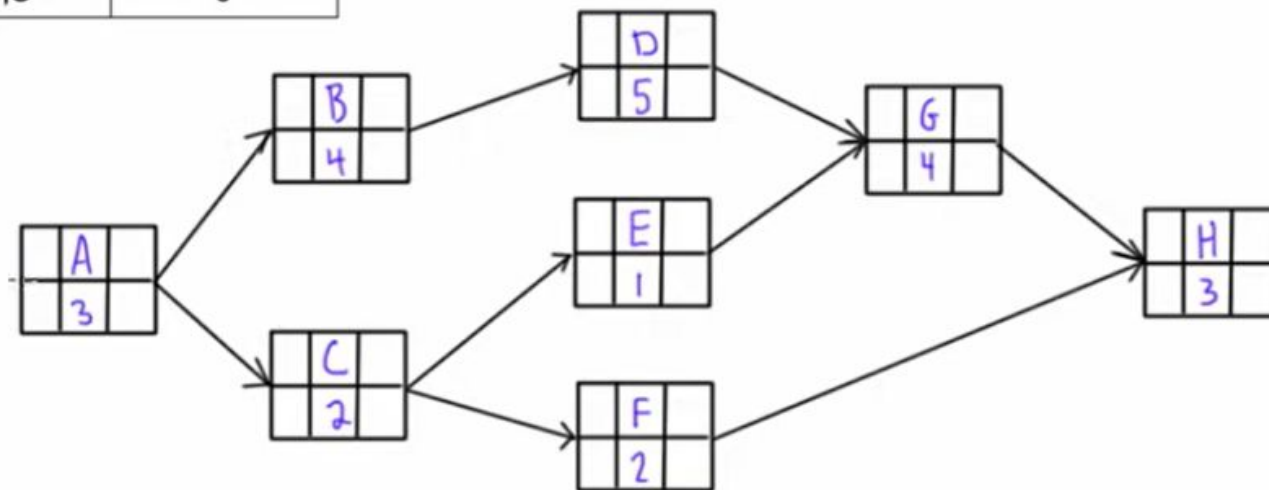
Activity	Predecessor	t_a	t_m	t_b	Mean (μ)	Variance (σ^2)	Standard Deviation (σ)
A	-	2	4	9	4.9	1.36	1.17
B	A	5	8	14	8.5	2.25	1.50
C	B	4	10	13	9.5	2.25	1.50
D	B	4	7	10	7	1.00	1.00
E	C	11	14	20	14.5	2.25	1.50
F	D	9	13	16	12.8	1.36	1.17
G	E,F	2	4	6	4	0.44	0.67



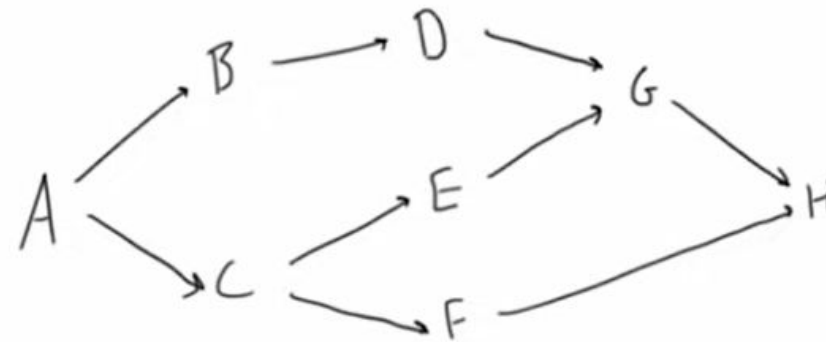
Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



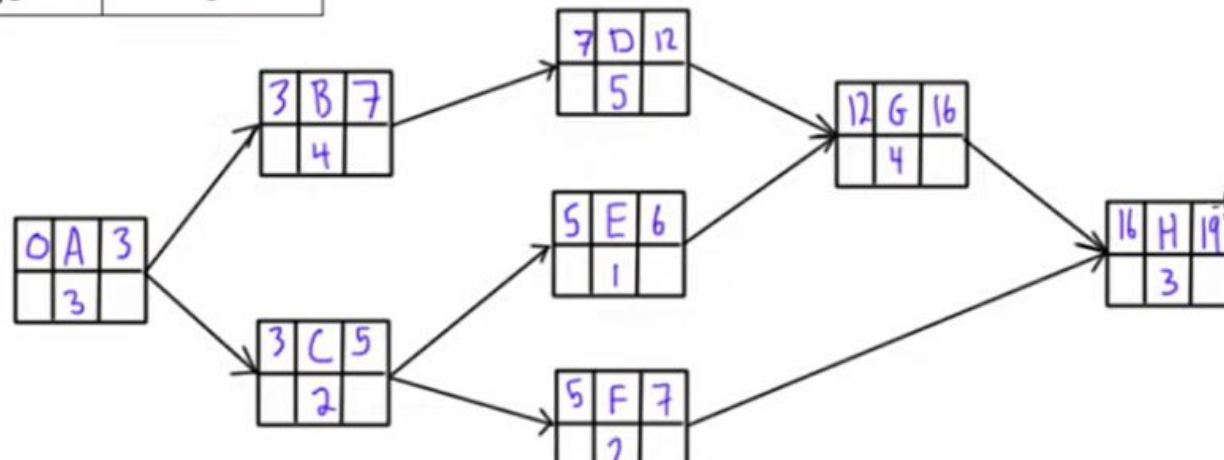
ES	Act	EF
LS	Dur	LF



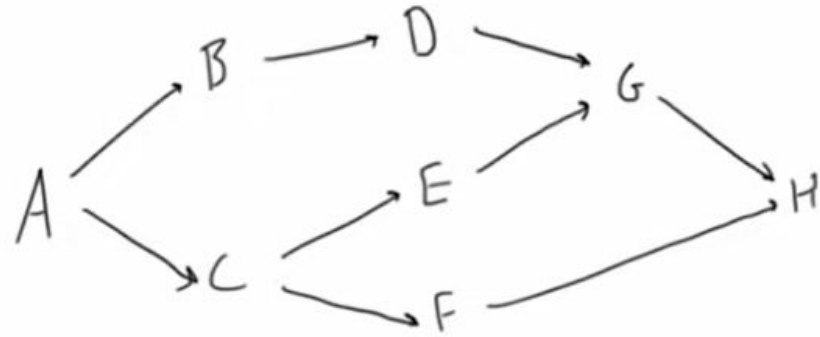
Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



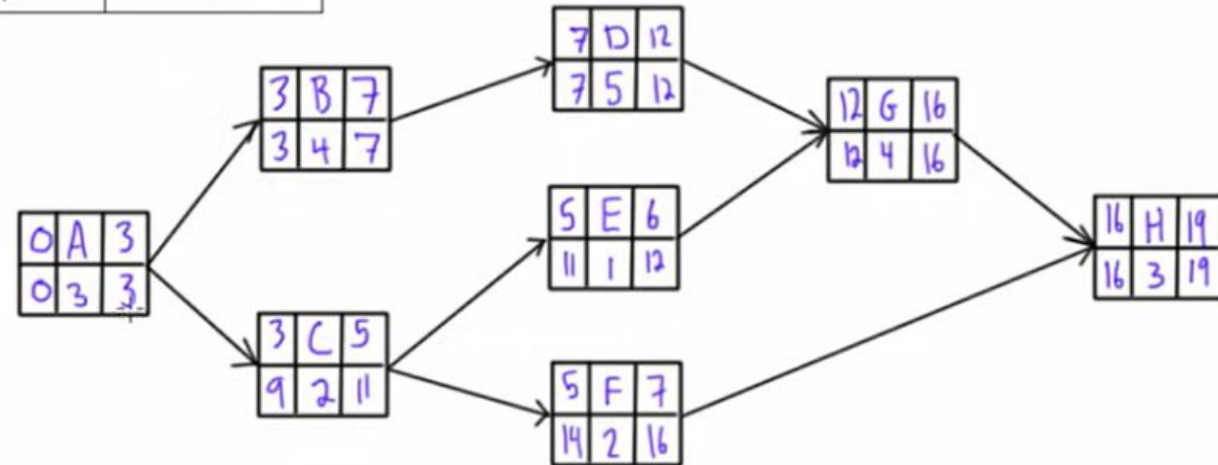
ES	Act	EF
LS	dur	LF



Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3

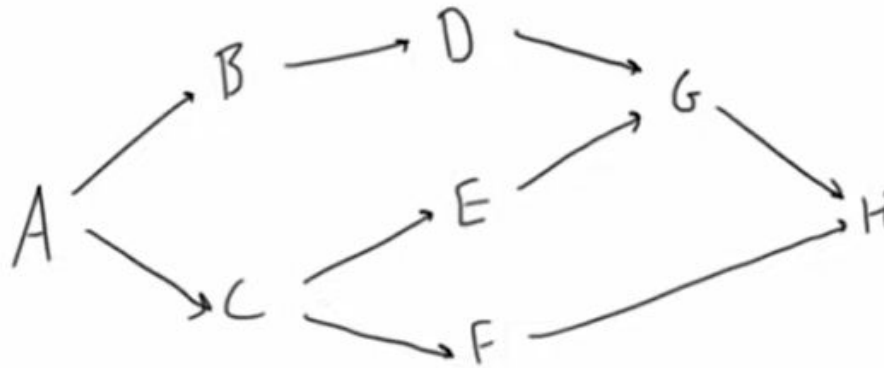


ES	Act	EF
LS	dur	LF

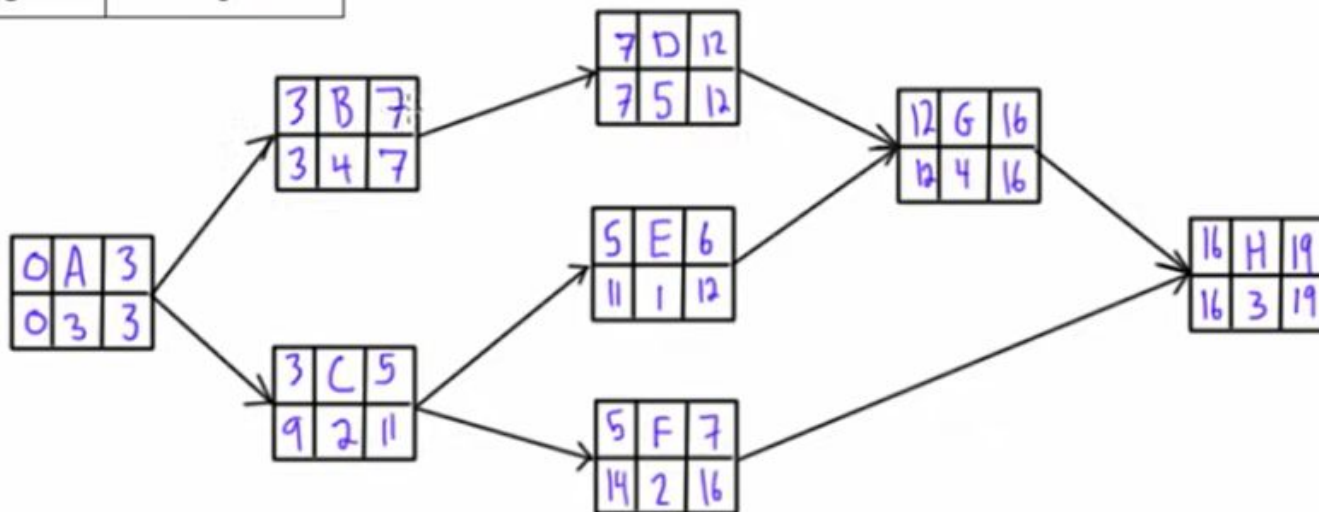


Example

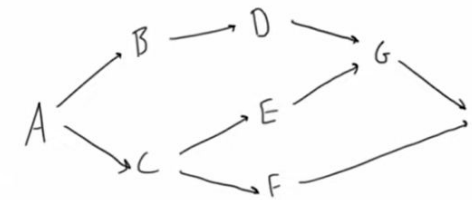
Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



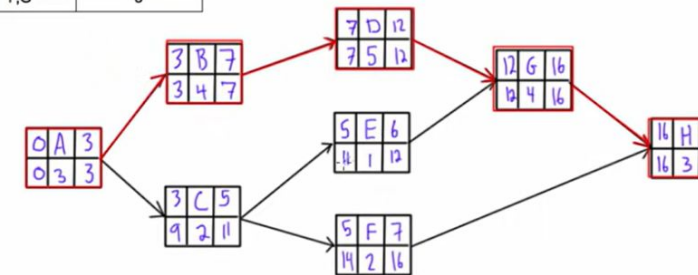
ES	Act	EF
LS	Dur	LF



Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



ES	Act	EF
LS	Dur	LF



Critical path is: A,B,D,G,H
Because TF and FF both are zero

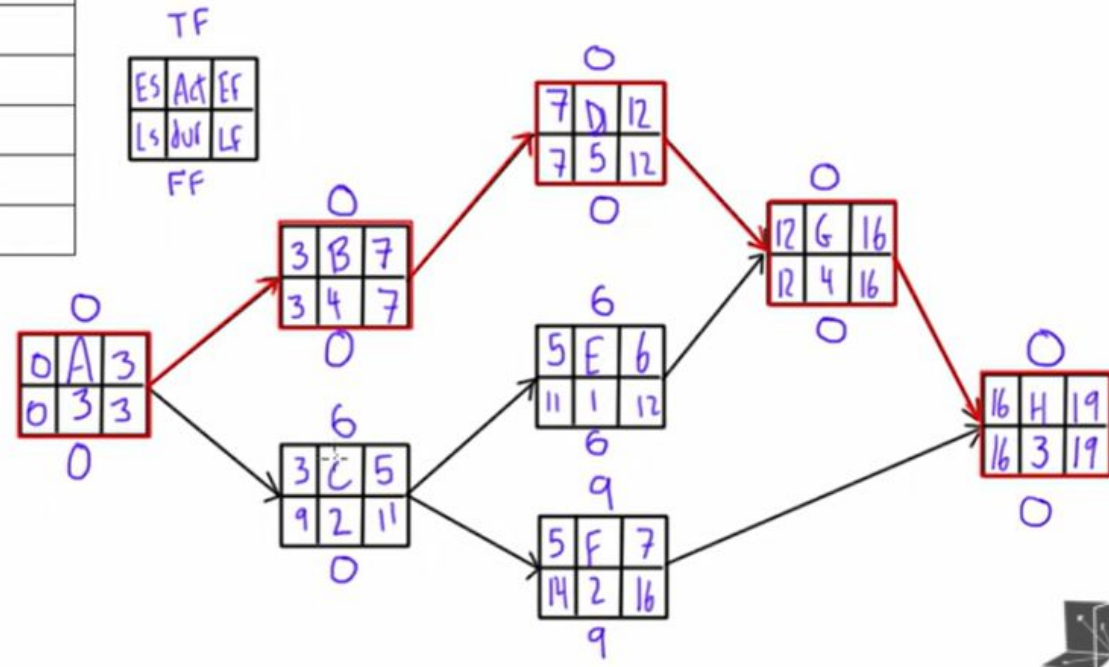
TF and FF

Total Float=TF

TF=LS-ES

FF =LF-EF

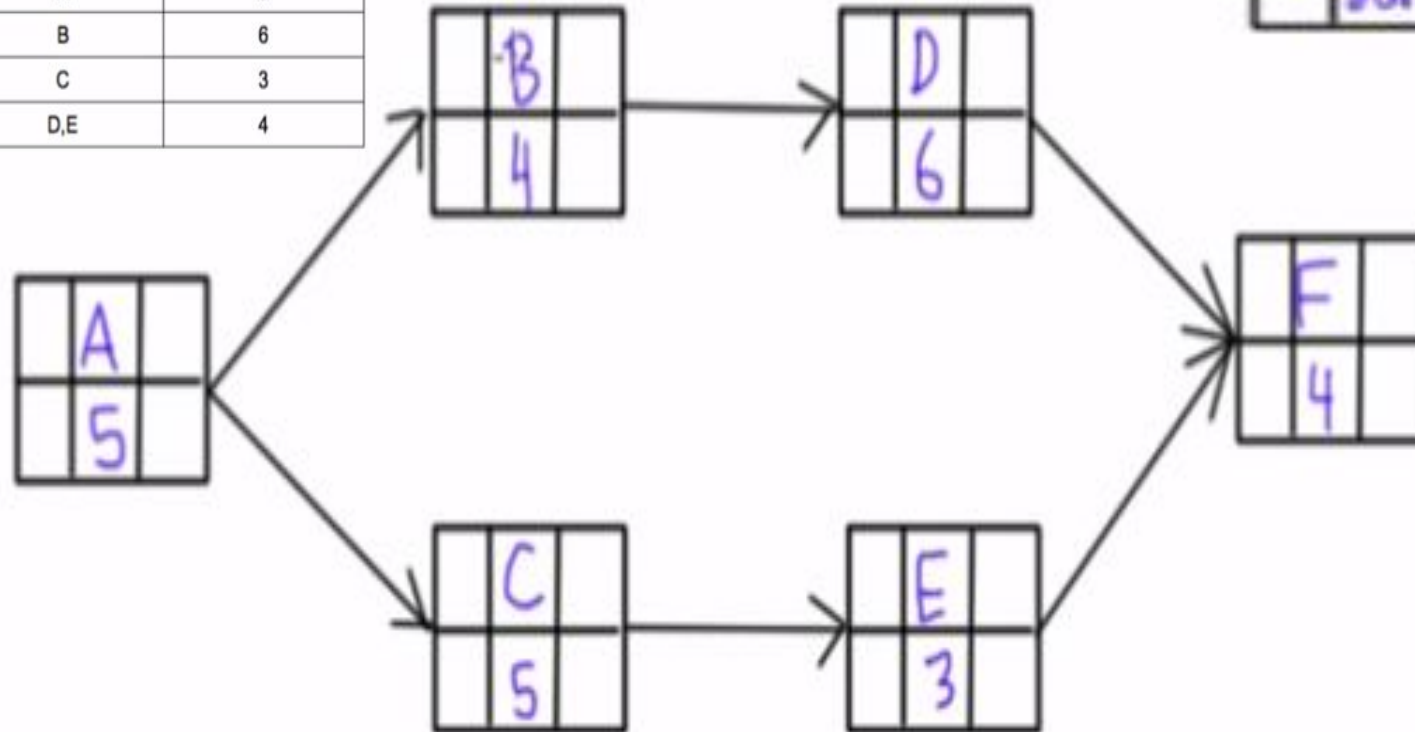
Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



Critical path is: A,B,D,G,H
Because TF and FF both are zero

Example: Activity Network

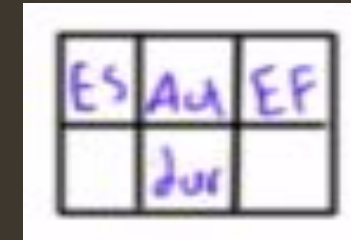
Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D,E	4



ES	AS	EF
	dur	

Example: Activity Network- Early Start and Early finish

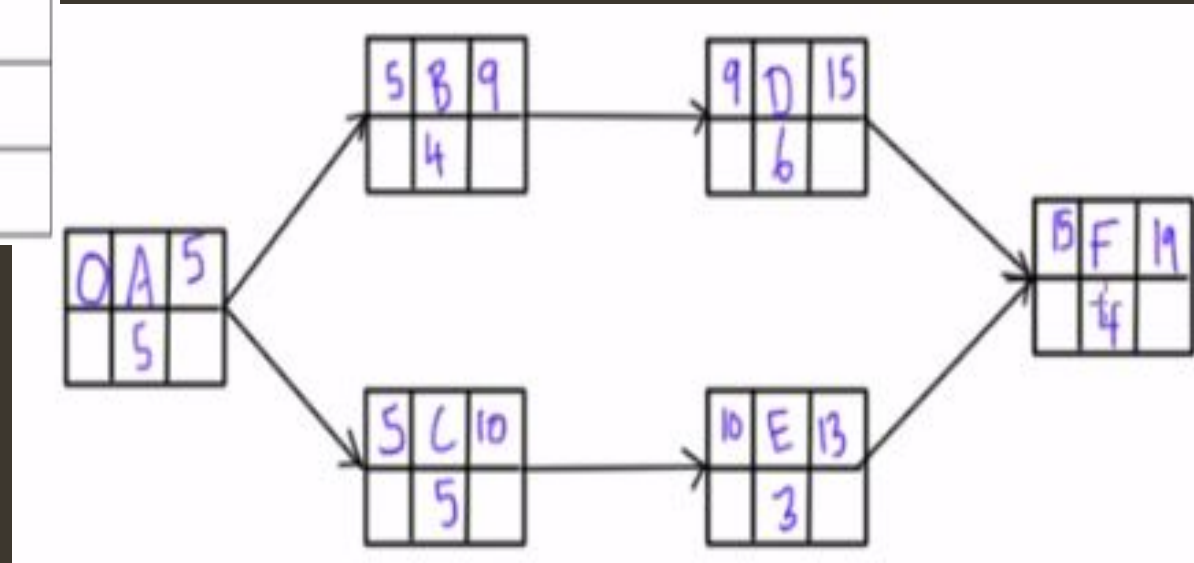
Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D,E	4



$A, B, D, F = 5 + 4 + 6 + 4 = 19$

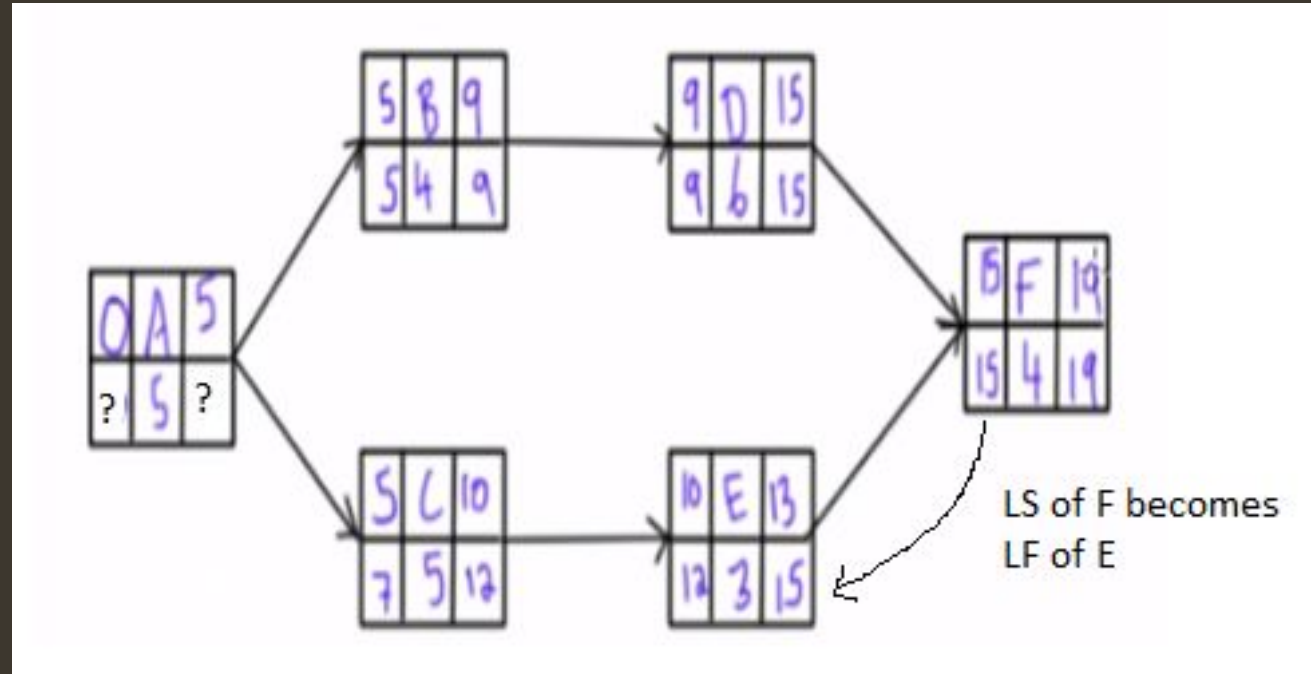
$A, C, E, F = 5 + 5 + 3 + 4 = 17$

Therefore, ABDF is critical path



Example: Activity Network- Late Start and Late finish

ES	AS	EF
LS	DS	LF



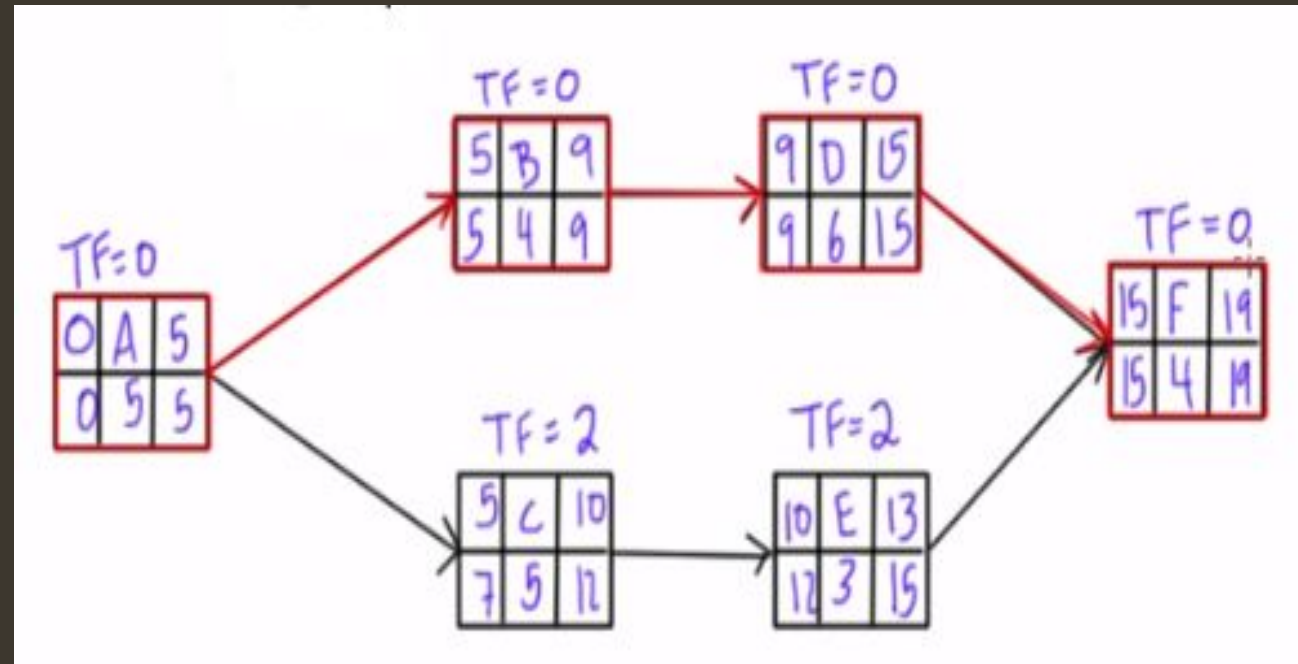
Example: Activity Network- Total Float(Slack)

Total float is how long an activity can be delayed, without delaying the project completion date. On a critical path, the total float is zero.



Total Float=TF

TF=LS-ES OR TF=LF-EF



Example: Activity Network- Free Float(Slack)

Free float is how long an activity can be delayed, without delaying the Early Start of its successor activity.

